Image Preprocessing Pipeline for Video Frame Enhancement

Final Group Project Report

Abstract

Image preprocessing is an essential component of computer vision systems, as images and video frames usually contain many types of noise, illumination, blur, and low contrast, etc., that may impact the image analysis and processing results. This project proposes an image preprocessing pipeline that is aimed at improving the quality of some video image frames. The image preprocessing pipeline has four stages: image noise removal by median filtering, image illumination correction by Contrast Limited Adaptive Histogram Equalization (CLAHE), image blur removal by image sharpening with quantitative sharpness evaluation, and image enhancement by Canny edge detection. The experimental results show that the proposed image preprocessing pipeline is effective for image quality improvement, so it can be applied for image analysis and processing.

1. Introduction

Introduction to image preprocessing highlights its crucial role in enhancing image quality prior to further analysis. Real-world video frames often suffer from issues such as noise, inconsistent lighting, blurriness, and low contrast, which, if not properly managed, can significantly hinder performance in later processing stages. This group project aims to develop a structured image preprocessing scheme, with each member assigned to implement a particular preprocessing stage, ultimately contributing to a comprehensive solution.

2 Dataset Description.

The data set used in this project consists of selected frames obtained from the video. The selected frames are stored in the form of PNG images and arranged in multiple folders within the main directory. Each folder contains selected frames for experimental data analysis. A portion of the selected frames is used to obtain the visual and quantitative evidence for the preprocessing techniques.

3.Methodology

This section describes the different preprocessing techniques used by the different members of the group using Python programming with the OpenCV library.

3.1 Noise Removal by Member 1:Unwanted variation in the intensity of the images was removed using the median filtering method, which removes salt and pepper noises without affecting the edges of the images. The images were converted to grayscale, and the median blur filter was used to remove the noises.

3.2 Illumination Correction by Member 2: To correct the illumination of the images, the Contrast Limited Adaptive Histogram Equalization (CLAHE) method was used to improve the contrast of the images by performing histogram equalization on the images, which reduces the effect of noises.

3.3 Blur Removal by Member 3:Unsharp images were removed using the unsharp masking method, which improves the clarity of the images by performing a focus test using the variance of the Laplacian, which gives the score of the clarity of the images.

3.4 Edge Enhancement by Member 4: The images were enhanced using the Canny edge detection method, which extracts the edges of the images, resulting in better image representation, which can be used for further analysis of the images.

4. Results and Discussion

The results obtained from the preprocessing pipeline show that all the techniques contribute to the enhancement of the image. Noise removal reduced the random fluctuations in the image, illumination correction enhanced the contrast and visibility of the image, blur reduction enhanced the sharpness of the image, and edge enhancement enhanced the important features in the image. Thus, all the techniques together provide an effective preprocessing pipeline that enhances the quality of the video frame. The use of visual and quantitative data enhances the validity of the results.

5. Conclusion

This group project has effectively implemented a four-stage image preprocessing pipeline to enhance the video frame. Each stage in the preprocessing pipeline effectively addresses the problem in the image. The results obtained show that the proposed pipeline is effective and can be used in computer vision.

6.Pipe Line

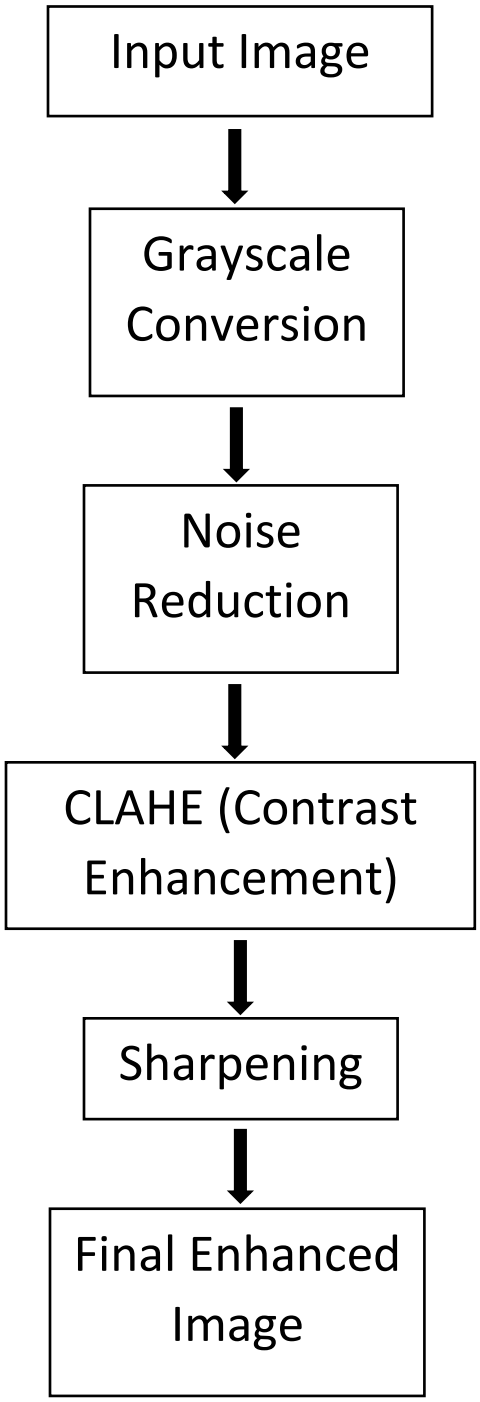
Input Image
Grayscale Conversion
Noise Reduction
CLAHE (Contrast Enhancement)
Sharpening
Final Enhanced Image

Figure 1: Original Input Frame

Figure 2: After Noise Reduction (Median Filter)

Figure 3: After Illumination Correction (CLAHE)
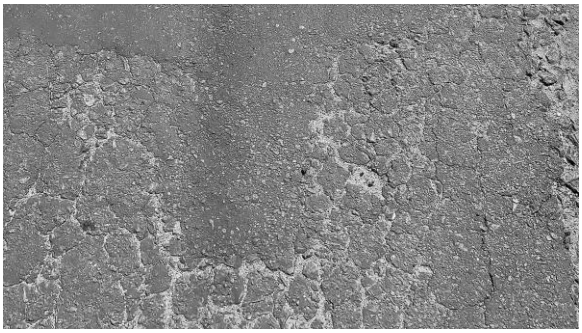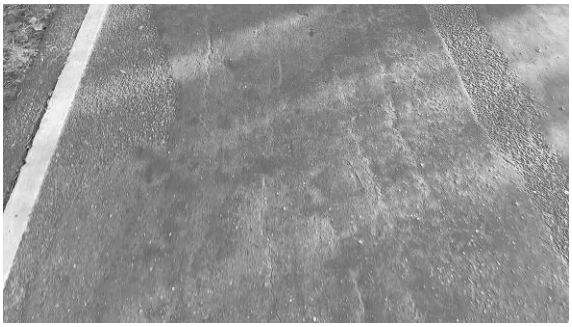
Figure 4: After Sharpening